

Impacts of recent climate warming, cultivar changes, and crop management on winter wheat phenology across the Loess Plateau of China

Liang He^{a,*}, Senthold Asseng^b, Gang Zhao^c, Dingrong Wu^d, Xiaoya Yang^e, Wei Zhuang^a, Ning Jin^f, Qiang Yu^{a,g}

^a Key Laboratory of Water Cycle and Related Land Surface Processes, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing 100049, China

^b Agricultural and Biological Engineering Department, University of Florida, Gainesville, FL 32611, USA

^c Crop Science Group, Institute of Crop Science and Resource Conservation (INRES), University of Bonn, Katzenburgweg 5, D-53115 Bonn, Germany

^d Chinese Academy of Meteorological Sciences, Beijing 100081, China

^e College of Applied Meteorology, Nanjing University of Information Science and Technology, Nanjing 210044, China

^f Shanxi Climate Center, Taiyuan 030002, China

^g Plant Functional Biology & Climate Change Cluster, University of Technology Sydney, PO Box 123, Broadway 2007, NSW, Australia

ARTICLE INFO

Article history:

Received 12 January 2014

Received in revised form 7 September 2014

Accepted 18 September 2014

Available online 15 October 2014

Keywords:

Winter wheat phenology

Climate warming

Cultivar change

Loess Plateau

APSIM

ABSTRACT

Crop yields are influenced by growing season length, which are determined by temperature and agronomic management, such as sowing date and changes in cultivars. It is essential to quantify the interaction between climate change and crop management on crop phenology to understand the adaptation of farming systems to climate change. Historical changes in winter wheat phenology have been observed across the Loess Plateau of China during 1981–2009. The observed dates of sowing, emergence, and beginning of winter dormancy were delayed by an average of 1.2, 1.3, and 1.2 days decade⁻¹, respectively. Conversely, the dates of green-up (regrowth after winter dormancy), anthesis, and maturity advanced by an average of 2.0, 3.7, and 3.1 days decade⁻¹, respectively. Additionally, the growth duration (sowing to maturity), overwintering period, and vegetative phase (sowing to anthesis) shortened by an average of 4.3, 3.1, and 5.0 days decade⁻¹, respectively. The changes in phenological stages and phases were significantly negatively correlated with a temperature increase during this time. Differently to most other phase changes, the reproductive phase (anthesis to maturity) prolonged by an average of 0.7 day decade⁻¹, but this was spatially variable. The prolonged reproductive phase was due to advanced anthesis dates and consequently caused the reproductive phase to occur during a cooler part of the season, which led to an extended reproductive phase. Applying a crop simulation model using a field-tested standard cultivar across locations and years indicated that the simulated phenological stages have accelerated with the warming trend more than the observed phenological stages. This indicated that, over the last decades, later sowing dates and the introduction of new cultivars with longer thermal time requirement have compensated for some of the increased temperature-induced changes in wheat phenology.

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1. Introduction

The Loess Plateau (LP) is located in the northwest China and covers an area of 0.65 million km² (Fig. A.1). Continental semi-arid and arid climate dominates this area with annual mean temperature from 3.6 to 14.3 °C and annual precipitation from 200 to 700 mm from the northwest to southeast. Most precipitation occurs in the

summer season, which causes serious soil erosion in poor vegetation cover areas. The crops in the LP mainly depend on rainfall because of the lack of water available for irrigation (He et al., 2014). As a result of the adverse environmental conditions such as soil erosion, unreliable rainfall and desertification, the LP is one of most vulnerable areas in China to climate change.

Wheat is the dominant crop in the Loess Plateau. The LP has a planting area of approximately 5 million ha, which accounts for 10% of the total wheat production in China (Wang and Li, 2010). During the last decades, air temperature has increased by 0.6 °C decade⁻¹ in this region (Wang X.C. et al., 2011; Wang Q. et al., 2012)

* Corresponding author. Tel.: +86 010 64856515.
E-mail address: xhl860728@163.com (L. He).

which is more than the global average temperature change ($0.13^{\circ}\text{Cdecade}^{-1}$) (IPCC, 2013). Increasing temperature accelerates the crop phenological development, which could potentially impact wheat production (Tao et al., 2014). A better understanding of how crop phenology responds to increasing temperature is essential for adapting the farming practices and crop breeding to mitigate the negative effects and even taking advantage of climate change (Hoffmann and Sgrò, 2011).

Crop phenology is codetermined by environmental conditions, mainly thermal time requirement, and agronomic practices, including sowing dates and cultivar characteristics. Detecting phenological responses to climate warming is difficult because of the continuously changing sowing dates and introducing new cultivars (Chmielewski et al., 2004; Estrella et al., 2007). For example, earlier flowering dates in winter wheat were observed throughout the last 70 years in the U.S. Great Plains because of a temperature increase during spring, which caused less time for light interception and biomass growth (Hu et al., 2005). In another study, Xiao et al. (2012) reported that from 1981 to 2009, climate warming in the North China Plain caused the dates of green-up after winter dormancy, anthesis, and maturity of winter wheat to occur in average of 1.1, 2.7, and 1.4 days earlier per decade, respectively. Wang et al. (2008) suggested that a warming trend in the LP from 1983 to 2004 led to earlier stem elongation, booting, anthesis, and maturity of winter wheat by an average of 13.2, 9.8, 11.0, and 10.8 days, respectively. However, there are uncertainties in this study because of the limited number of stations (only one station was used) and the cultivar shift during this period (Liu Y. et al., 2010; Wang et al., 2013). Warming temperature could accelerate the crop development, while a long-duration cultivar or late sowing date could result in a delayed development. Statistical models could fail to capture the interactions between environmental changes, management practices and the cultivar change. Process-based crop models can dynamically simulate the interactions between crops, environmental conditions and management practices (Zhao et al., 2014). The effect of one factor can be separated from other factors (Liu Y. et al., 2010; Liu L. et al., 2012; Wang et al., 2013).

This study examined the trends of winter wheat phenological development using observed data from 16 agro-meteorological stations in the LP from 1981 to 2009. We correlated these records with temperature trends to analyze how climate warming has affected winter wheat phenological stages and phases. A crop model was applied to separate the effects of climate warming, crop management and cultivar changes on phenology.

2. Material and methods

2.1. Agro-meteorological stations and data

Winter wheat is mainly distributed across the southeastern and central LP. In this study, we selected 16 stations with the largest number (24 or more) of recorded years (Fig. A.1). The data on winter wheat phenology from 1981 to 2009 are from local agro-meteorological experimental stations maintained by the Chinese Meteorological Administration (CMA). Phenological data included dates of sowing, emergence, start of dormancy, green-up (start of regrowth in spring after dormancy), anthesis (50% anthesis), and maturity (physiological maturity). With these data, four phenological phases were derived, including growth duration (sowing to maturity), overwintering period (DG, start of dormancy to green-up), vegetative growth stage (SA, sowing to anthesis), and reproductive growth stage (AM, anthesis to maturity). The local farmers determined the management practices of the winter wheat crop. Approximately every 3–6 years (average 10 cultivars per station), these farmers changed to a new wheat cultivar, which

could alter crop temperature requirements. Cultivar names were listed in Table A.1. Historical daily weather data, including maximum, minimum, and mean temperatures for 1980–2009 at the 16 stations, were also obtained from CMA website at <http://cdc.cma.gov.cn/>.

2.2. Data analysis

Trends in observed phenology and corresponding mean temperature were estimated with a linear regression, using the year as the independent variable. Time windows for calculating temperature trends were determined by the longest phenological stages at each station. For example, the time window of growth duration (i.e. sowing to maturity) was from the earliest sowing date to the latest date of maturity during the last decades at each station. By holding a time window of a growth duration constant, the calculated temperature trend was independent from the corresponding phenology changes.

We correlated the sowing dates with monthly mean temperature during the month of occurrence of sowing to evaluate whether the sowing dates were driven by temperature. The responses of phenology to temperature were calculated as follows:

$$OP_{nt} = a_{nt}T_{nt} + b_{nt} + \epsilon_{nt} \quad (1)$$

where OP_{nt} is the observed phenological phases (DOY, day of year) or phenological length (days) for the n th stations in year t . T_{nt} is the mean daily temperature ($^{\circ}\text{C}$) during the corresponding development stage for the n th stations in year t . The variable a_{nt} is the coefficient of phenology responses to temperature ($\text{days}^{\circ}\text{C}^{-1}$) for the n th station. The variable b_{nt} represents the intercept, and ϵ_{nt} is the error term for each station. The regression coefficients are considered as the phenology responses to temperature and crop management changes.

2.3. Thermal time calculation and phenology simulation with the APSIM-wheat model

The Agricultural Production Systems sIMulator (APSIM) is a modular modeling framework developed by the Agricultural Production Systems Research Unit in Australia (Keating et al., 2003). In the APSIM model, wheat phenology is determined by accumulation of thermal time (ATT in degree-days) and the cultivar specific thermal time requirement for each development stage. The phenology before flowering is also influenced by the vernalization and photoperiod sensitivity. The total thermal time from sowing to anthesis (SA) and from anthesis to maturity (AM) is calculated as:

$$ATT = \sum_{i=1}^n DTT \quad (2)$$

where DTT is thermal time per day, n is the days of phenological stages. DTT is calculated from 3-hourly air temperature interpolated from the daily maximum and minimum crown temperature. The detailed description of APSIM can be found at <http://www.apsim.info>.

APSIM operates at a daily time-step, and it is driven by daily weather data including daily maximum and minimum temperatures, rainfall, and solar radiation. In this study, APSIM version 7.4 was used.

We simulated the crop phenology by using a single cultivar and the same agronomic practices throughout the years to separate the effects of temperature, crop management, and cultivar on crop phenology. We simulated the anthesis, maturity date, and growth duration from 1981 to 2009 across all agro-meteorological stations in the LP. APSIM was calibrated with a single, most frequently-used cultivar (1981–1983) at each site. Sixteen cultivars, each different

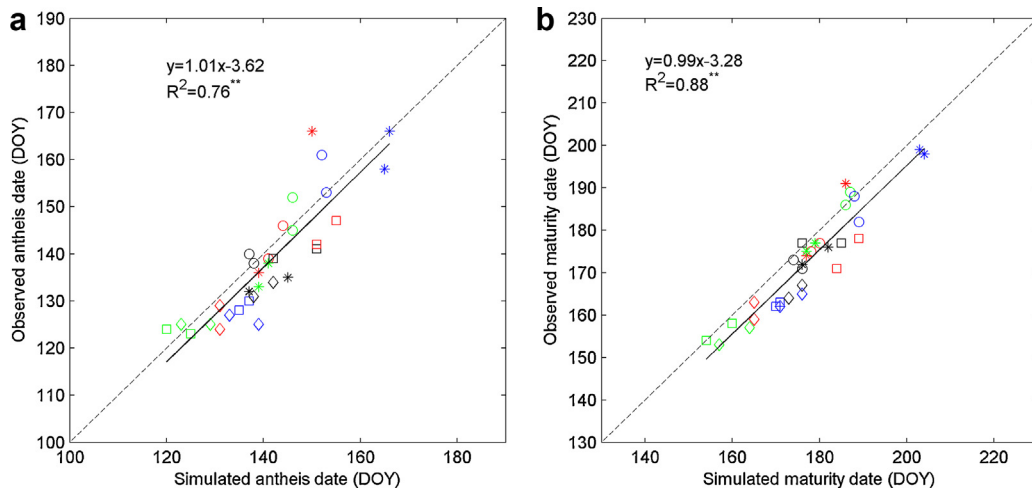


Fig. 1. Comparison of observed and simulated anthesis and maturity dates during the period of 1984–1985 at 16 stations. Different markers represent different stations.

by location, were used in the calibration. The model was then validated with phenology observations from 1984 to 1985 and used to simulate wheat phenology for 1981–2009, using the same cultivar and management every year. The response of simulated phenology (growth duration, sowing to anthesis, anthesis to maturity) to temperature was calculated as the following:

$$SP_{nt} = c_{nt}T_{nt} + d_{nt} + \epsilon_{nt} \quad (3)$$

where SP_{nt} is the simulated phenological length (days) for the n th stations in year t . T_{nt} is the average phenological phase temperature ($^{\circ}\text{C}$) for the n th stations in year t . The variable c_{nt} is the coefficient of phenology responses to temperature ($\text{days } ^{\circ}\text{C}^{-1}$) for the n th station. The variable d_{nt} represents the intercept, and ϵ_{nt} is the error term for each station.

Fig. 1 shows the comparison of observed and simulated anthesis and maturity dates during validation years (1984–1985) at 16 stations. Simulated anthesis (slope = 1.01, $R^2 = 0.76$, $p < 0.01$) and maturity (slope = 0.99, $R^2 = 0.88$, $p < 0.01$) dates agreed well with the observations, suggesting that the APSIM performed well for this region.

2.4. Difference between observed and simulated responses to temperature

The regression coefficient (a_{nt}) in Eq. (1) reflects the phenology response to temperature, sowing dates and cultivar change. The regression coefficient (c_{nt}) in Eq. (3) only includes the influence of temperature. A positive difference of the regression coefficients ($a_{nt} - c_{nt}$) indicates farmers changed to longer-duration cultivars, while a negative one indicates a change to shorter-duration cultivars introduced over the last decades. The differences in regression coefficients were tested by paired t -test for each stage.

3. Results

3.1. Temperature trends

There was a general warming trend between 0.4 and $0.9^{\circ}\text{C decade}^{-1}$ during sowing to maturity (Fig. 2a), between 0.2 and $0.9^{\circ}\text{C decade}^{-1}$ during winter dormancy (Fig. 2b), between 0.4 and $0.9^{\circ}\text{C decade}^{-1}$ during sowing to anthesis (Fig. 2c), and between 0.2 and $1.1^{\circ}\text{C decade}^{-1}$ during anthesis to maturity (Fig. 2d).

3.2. Spatial and temporal variability of winter wheat phenology stages

Winter wheat in the LP is generally sown in late September to early October (Table 1). The spatial trends of sowing dates are shown in Fig. 3a. In 1981–2009, the sowing dates were delayed on average by $1.2 \text{ days decade}^{-1}$. A delay occurred at 12 stations (16 sites in total), five of those statistically significantly ($p < 0.05$). However, the sowing dates at four stations were advanced, but only two of those were advanced significantly. The trends of emergence dates were similar to sowing dates and were likely directly related to delayed sowing. The average delay in emergence date was $1.3 \text{ days decade}^{-1}$ (Fig. 3b).

Winter wheat in the LP starts winter dormancy at the beginning of December and greens up (end of dormancy) at late February to early March in the following year (Table 1). The dates of dormancy (start of dormancy) were delayed at 10 stations (Fig. 3c). The delayed trends were significant at two eastern stations (Fig. 3c) on average of $1.2 \text{ days decade}^{-1}$. Green-up dates advanced at 11 stations (Fig. 3d), ranging from -1.2 to $-8.7 \text{ days decade}^{-1}$ (four stations significantly). However, green-up dates at five stations were delayed (only one significantly). The dates of green-up have advanced by an average of $2.0 \text{ days decade}^{-1}$.

Anthesis and maturity of winter wheat in the LP generally occurs in May and June, respectively (Table 1). The anthesis dates at all stations were advanced (fourteen of them were significant), ranging from -1.8 to $-5.9 \text{ days decade}^{-1}$ (Fig. 3e). Similarly, the maturity dates were significantly advanced at 11 stations (Fig. 3f). Anthesis and maturity dates advanced by an average of 3.7 and $3.1 \text{ days decade}^{-1}$, respectively.

3.3. Spatial and temporal variability of winter wheat phenology phases

The delay of sowing ($1.2 \text{ day decade}^{-1}$, Fig. 3a) and advancement of maturity ($-3.1 \text{ day decade}^{-1}$, Fig. 3f) led to a reduction of the growth duration from sowing to maturity by $-4.3 \text{ days decade}^{-1}$ (Fig. 4a). The growth durations at 9 stations were significantly decreased and ranged from -4.0 to $-10.5 \text{ days decade}^{-1}$. The winter dormancy period (DG in Fig. 4b) decreased by an average of $3.1 \text{ days decade}^{-1}$ from 1981 to 2009, but significantly at only 6 stations. One station in the southwest LP had a significantly increased DG (Fig. 4b).

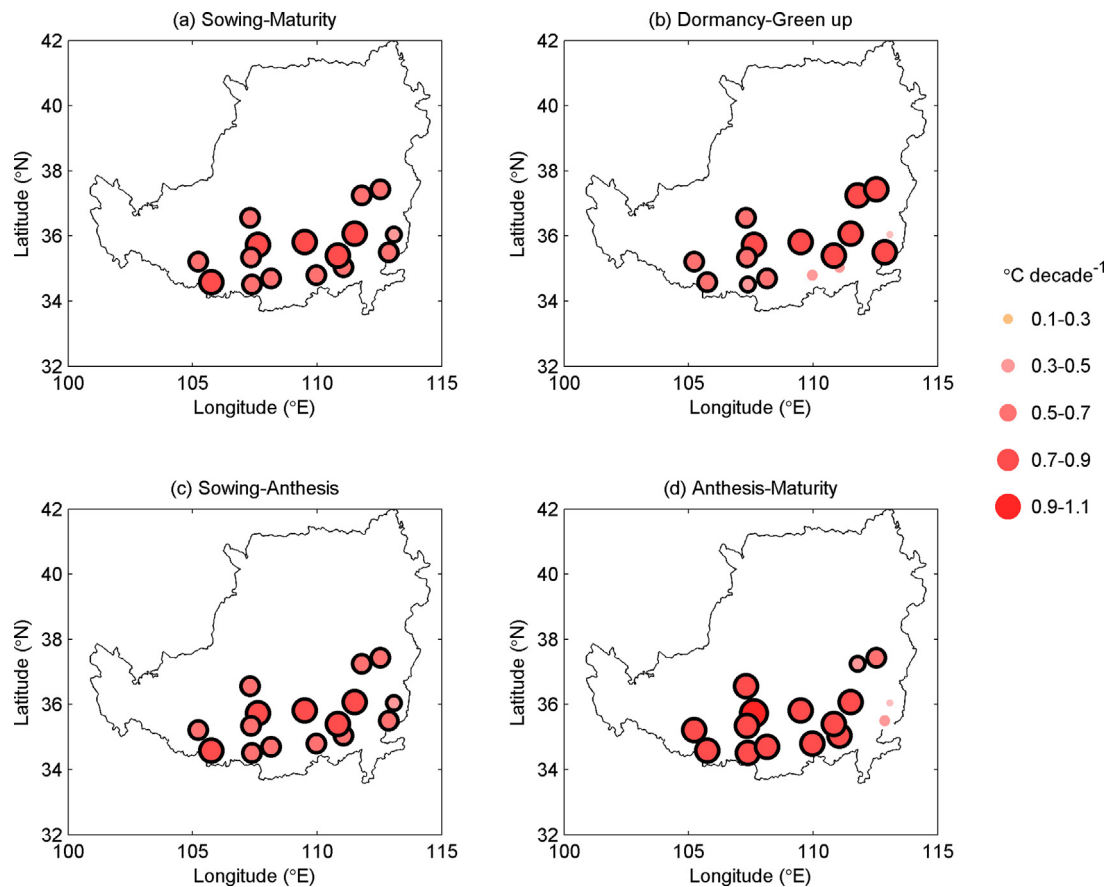


Fig. 2. Observed trends in mean temperature during the phenological phases in the Loess Plateau from 1981 to 2009. (a) Sowing–maturity; (b) dormancy–green up; (c) sowing–anthesis; (d) anthesis–maturity.

The decrease of growth duration (-4.3 days decade $^{-1}$, Fig. 4a) was caused by a reduction of SA (sowing to anthesis). SA decreased at 15 stations (significantly at 11 stations) by an average of -5.0 days decade $^{-1}$ (Fig. 4c). In contrast, the reproductive growth stage and the phase of AM (anthesis to maturity) significantly increased at four eastern stations (Fig. 4d) by an average of 0.7 days decade $^{-1}$. The AM at five western stations and an eastern station decreased, but not significantly.

3.4. Spatial and temporal variability of winter wheat cultivar thermal characteristics

The thermal requirements of cultivars (Fig. 5) from sowing to anthesis increased at 12 stations by an average of 25.9 °C d decade $^{-1}$, ranging from 39.5 to 81.8 °C d decade $^{-1}$ (significantly at six stations). Similarly, the thermal time of cultivars during the reproductive stage (anthesis to maturity) increased at

14 stations by an average of 22.6 °C d decade $^{-1}$, ranging from 28.0 to 77.1 °C d decade $^{-1}$ (significant at 7 stations).

3.5. Responses of observed phenology to temperature

Sowing dates were positively correlated with temperature at 10 stations (only one significantly; see Table 2 and Fig. 6). The start of winter dormancy was positively correlated with temperature at 15 stations (two stations significantly). The sowing dates and start of dormancy were delayed by an average of 0.88 and 2.27 days °C $^{-1}$, respectively. Subsequent phenological stages such as green-up, anthesis, and maturity had a significant negative correlation with temperature (see Table 2 and Fig. 6). On average, the phenology was reduced by -4.03 days °C $^{-1}$ for green-up dates, -3.72 days °C $^{-1}$ for anthesis dates, and -2.94 days °C $^{-1}$ for maturity dates. The growth duration from sowing to maturity, sowing to anthesis, and anthesis to maturity were reduced by an average of -5.29 , -5.88 , and -0.75 days °C $^{-1}$, respectively.

Table 1
Average observed phenology of winter wheat in the Loess Plateau during the period of 1981–2009.

Phenology	Sowing	Emergence	Dormancy	Green-up ^a	Anthesis ^b	Maturity ^c
Average date (day of year)	273	282	338	61	135	169
Standard deviation	5.8	6.4	7.9	11.7	10.0	10.8

^a Start re-growth in spring after dormancy.

^b 50% anthesis.

^c Physiological maturity.

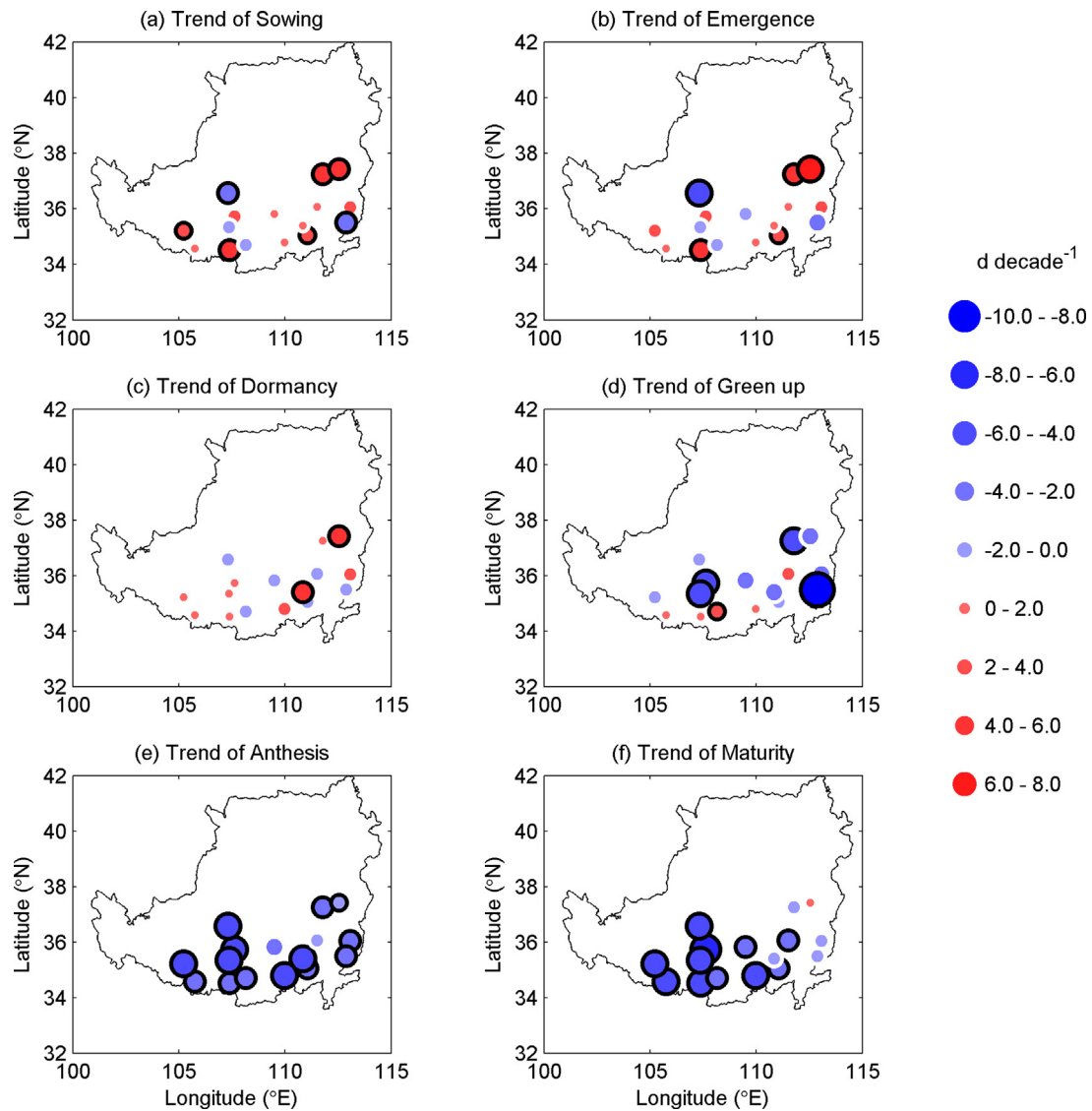


Fig. 3. Observed trends in phenology of winter wheat in the Loess Plateau from 1981 to 2009. (a) Sowing date; (b) emergence date; (c) start of dormancy; (d) green-up date (start of re-growth after dormancy in spring); (e) anthesis (50% anthesis) date; (f) maturity (physiological maturity) date; black circles indicate statistical significant trends at $p = 0.05$ levels.

3.6. Responses of simulated phenology to temperature

The response of three simulated phenological stages to temperature is shown in Table 3 and Fig. 7. Similar to the observations,

there was also a negative correlation between simulated phenological stages and temperature. Significant negative correlations occurred at all stations for the growth duration from sowing to maturity and sowing to anthesis and at nine stations for anthesis

Table 2
Summary of observed phenology responses to temperature for winter wheat in the Loess Plateau for 1981–2009.

Phenology	No. neg. ^a	No. pos. ^b	No. sig. neg. ^c	No. sig. pos. ^d	Reg.mean ^e (days °C ⁻¹)
Sowing	6	10	0	1	0.88
Dormancy	1	15	0	2	2.27
Green-up	15	1	11	0	-4.03
Anthesis	16	0	15	0	-3.72
Maturity	16	0	15	0	-2.94
Growth duration ^f	15	1	10	0	-5.29
Sowing-Anthesis	15	1	11	0	-5.88
Anthesis-Maturity	12	4	4	0	-0.75

^a Number of locations with negative regression coefficients.

^b Number of locations with positive regression coefficients.

^c Number of locations with significant negative regression coefficients.

^d Number of locations with significant positive regression coefficients.

^e Mean of regression coefficients.

^f Duration of sowing to maturity.

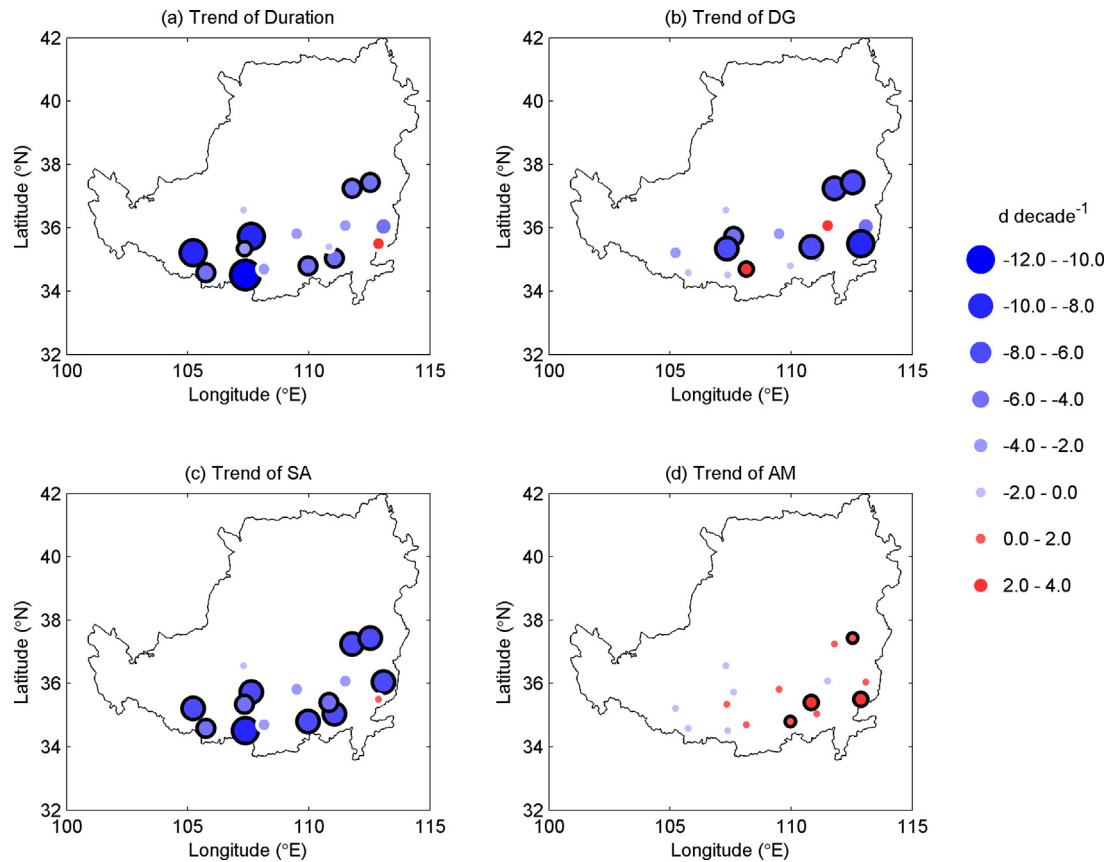


Fig. 4. Observed trends in phenology of winter wheat in the Loess Plateau from 1981 to 2009. (a) Total growth duration from sowing to maturity; (b) DG (dormancy to green-up); (c) SA (sowing to anthesis) and (d) AM (anthesis to maturity); Black circles indicate statistical significant trends at $p = 0.05$ levels.

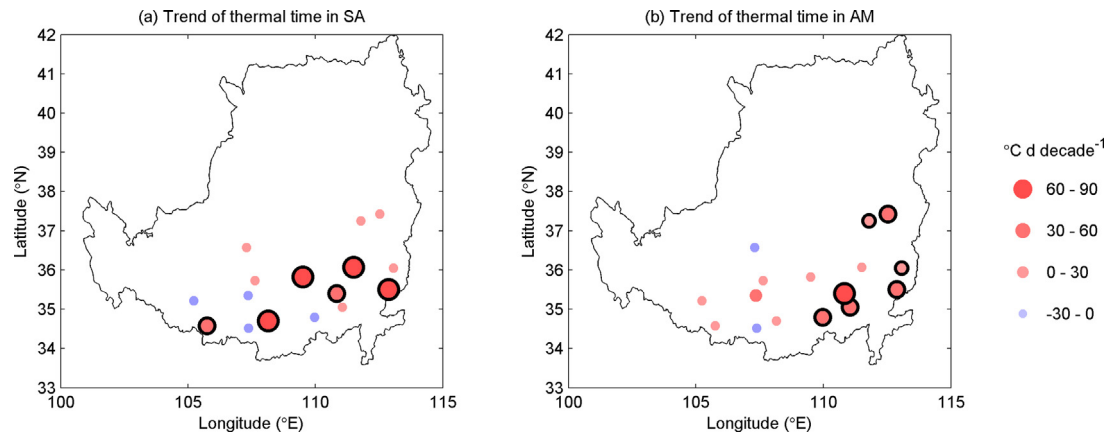


Fig. 5. Observed trends in thermal time required for (a) sowing to anthesis and (b) anthesis to maturity. Black circles indicate statistical significant trends at $p = 0.05$ levels.

Table 3
Summary of simulated phenology phases responses to temperature increase for winter wheat in the Loess Plateau for 1981–2009.

Phenology	No. neg. ^a	No. pos. ^b	No. sig. neg. ^c	No. sig. pos. ^d	Reg.mean ^e (days °C ⁻¹)
Growth duration ^f	16	0	16	0	−8.09
Sowing–Anthesis	16	0	16	0	−8.28
Anthesis–Maturity	16	0	9	0	−0.71

^a Number of locations with negative regression coefficients.
^b Number of locations with positive regression coefficients.
^c Number of locations with significant negative regression coefficients.
^d Number of locations with significant positive regression coefficients.
^e Mean of regression coefficients.
^f Duration of sowing to maturity.

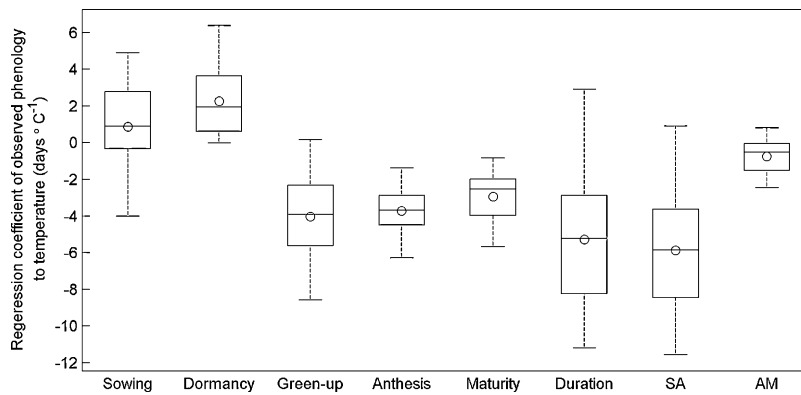


Fig. 6. Observed phenology changes versus temperature trends for winter wheat in the Loess Plateau from 1981 to 2009. (SA: sowing–anthesis; AM: anthesis–maturity).

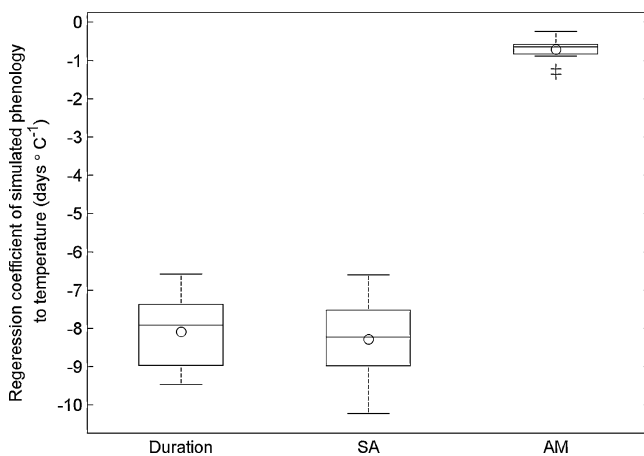


Fig. 7. Simulated changes of phenological phases versus observed temperature trends using a standard winter wheat cultivar at each location in the Loess Plateau from 1981 to 2009. (SA: sowing–anthesis; AM: anthesis–maturity).

to maturity. The mean simulated reductions in phases were by an average of 8.09 and 8.28 days °C⁻¹ for growth duration and sowing to anthesis, respectively, which were longer than the anthesis to maturity phase (0.71 days °C⁻¹).

3.7. Observed and simulated phenology

The temperature sensitivity of phenology for the observed data was less than the simulated results (Table 4), with statistically significant differences of 2.80 days °C⁻¹ for the growth duration from sowing to maturity, 2.40 days °C⁻¹ for sowing to anthesis, but higher than the simulated data with an insignificant difference of -0.04 °C⁻¹ for anthesis to maturity. The difference in observed and simulated changes in phenology with increased temperature suggested that some of the new cultivars introduced during the last decades had altered thermal time requirements.

4. Discussion

The observed changes in phenology for winter wheat across the Loess Plateau were likely caused by temperature increase over the recent decades, but were also partially caused by changes in sowing dates and the introduction of new cultivars with altered temperature requirements. The date of sowing is determined by farming management decisions, which can change in response to changes

in climate (Estrella et al., 2007). The change of sowing dates in the Loess Plateau is an adaptation response to the warming trend in this region. While soil moisture could also influence the sowing date decision, relative high and reliable rainfall during the month of sowing make this an unlikely change factor in this region. Winter wheat requires a short period of growth before winter dormancy. Too much pre-dormancy growth can negatively impact the survival rate of a crop during the winter dormancy. A warming trend over the recent decades provided additional suitable growing conditions before winter dormancy, which led farmers to postpone sowing dates accordingly. An increase in temperature of one degree resulted in an average of 0.88 days delay of sowing dates (Table 2). A similar trend in delaying the sowing date of winter wheat because of climate warming has been reported (1.5 days decade⁻¹) for the North China Plain (Xiao et al., 2012).

The observed warming trend in the LP caused a delay of the onset of winter dormancy (dormancy is mainly influenced by temperature) and advanced the green-up after winter dormancy, resulting in a reduced winter dormancy period (-3.1 days decade⁻¹). Similar results were also reported in the North China Plain (Xiao et al., 2012), where the winter dormancy period declined by -2.5 days decade⁻¹. Interestingly, the warming trend in winter can reduce mortality rates of winter wheat due to cold or freezing stress during the overwintering period (Gholipour, 2012; Nicholls, 1997; Wang et al., 2008).

The warming trend in the LP advanced anthesis and maturity dates and caused a general reduction of the durations from sowing to anthesis and from anthesis to maturity. An advancement of anthesis dates has been reported in several natural systems with recent climate warming (Abu-Asab et al., 2001; Fitter and Fitter, 2002; Sparks et al., 2000). Similarly, anthesis dates have been observed earlier for other agricultural crops (Hu et al., 2005; Sparks et al., 2005; Wang et al., 2008; Williams and Abberton, 2004; Xiao et al., 2012). Reductions in growing periods were also reported from field observations (Tao et al., 2012; Xiao et al., 2012; Zhang et al., 2013) and simulation studies (Sadras and Monzon, 2006). In comparison with previous analysis on winter wheat phenology, our finding (an average of 3.7 days decade⁻¹ earlier anthesis dates) is higher than the averages reported in the North China Plain (2.7 days decade⁻¹) (Xiao et al., 2012) and in the U.S. Great Plains (0.8–1.8 days decade⁻¹) (Hu et al., 2005), but less than the average reported for northwest China (11.0 days decade⁻¹) (Wang et al., 2008). Some of the phenology trend differences can be attributed to the difference in observed temperature trends. The temperature trend (0.4–0.9 °C decade⁻¹) in the LP was higher than the temperature trend in the North China Plain (0.2 °C decade⁻¹) (Fu et al., 2009) and the U.S. Great Plains (0.03–0.16 °C decade⁻¹) (Hu et al., 2005).

Table 4

Comparison of the responses of the phenological phases with mean temperature using the observed and simulated data in the Loess Plateau for 1981–2009.

Phenological stage	Regression coefficient using observed data (days °C ⁻¹) Reg.mean ^a	Regression coefficient using simulated data (days °C ⁻¹) Reg.mean ^a	Difference between observed and simulated data (days °C ⁻¹)	t-Test (p-value)
Growth duration ^b	-5.29	-8.09	2.80	0.0037**
Sowing–anthesis	-5.88	-8.28	2.40	0.0099**
Anthesis–maturity	-0.75	-0.71	-0.04	0.8881

^a Mean of regression coefficients.^b Duration of sowing to maturity

** Significant at the 0.01 probability level.

There is a small temperature trend difference compared with a site in northwest China ($0.5\text{--}0.9\text{ }^{\circ}\text{C decade}^{-1}$) (Wang et al., 2008), which reported a large difference in observed shortening of growth duration ($11.0\text{ days decade}^{-1}$). The large difference could be a result of the shortening trend in this study was an average across 16 stations over wheat growing regions in the LP, while the conclusion of Wang et al. (2008) was based on observations from a single station.

New cultivars are continuously being bred and introduced that are optimized for current growing conditions and adapted to the changing climate. These new cultivars could have new phenology characteristics. In our study, we separated the effects of new introduced cultivars in the LP on phenology by simulating the phenology of winter wheat with a crop model and using the same cultivar throughout the time period. The response to temperature in the simulation was higher than observed, suggesting that about 1/3 ($31.8\% = (2.8/8.09 + 2.4/8.28)/2$, Table 4) of the direct temperature impact on observed phenology was countered in the LP with new introduced cultivars. This means that an introduction of later-flowering and later-maturing cultivars during 1981–2009 in the LP partially compensated for the increased temperature effect on wheat phenology. A similar trend of using cultivars adapted to increasing temperature was reported for winter wheat in the North China Plain (Liu Y. et al., 2010), rice in China (Tao et al., 2013) and corn in the U.S. corn belt (Sacks and Kucharik, 2011). Conversely, in addition to the warming effect in China, a short-duration, early-flowering late rice cultivar also further accelerated phenology to avoid terminal drought risk (Zhang et al., 2013). A shorter-duration cultivar is likely to result in yield reductions, particularly in high input crops, because there is less time for biomass accumulation during the vegetative phase. Hence, farmers mostly introduced longer-duration cultivars in the LP to counteract the negative impacts of temperature trend on wheat phenology.

Phenology plays an important role in the formation of crop yield. Earlier anthesis could result in less biomass accumulation during the vegetative phase, resulting in reduced yields. However, earlier anthesis could also result in a longer anthesis to maturity phase, as observed in the LP ($0.7\text{ days decade}^{-1}$) because grain filling phase was advanced into an earlier and cooler period of the year. This could result in a higher grain yield (Ainsworth and Ort, 2010; Tao et al., 2006; Xiao et al., 2012). The phenological phase from anthesis to maturity has increased rather than decreased under the warming climate. One reason could be that the later cultivars had longer thermal time requirements than the earlier ones. Another reason could be that the advanced anthesis dates move this phase into an earlier and relatively cooler part of the year.

The climate is expected to warm faster in the future than in past decades (IPCC, 2013). The temperature in the LP is projected to increase by approximately $2.1\text{--}2.6\text{ }^{\circ}\text{C}$ (RCP 4.5 scenario of 19 GCMs) by 2050 (IPCC, 2013). Additionally, extreme events like droughts have become more frequent across the LP (Turner et al., 2011). Crop phenology would be accelerated with increasing temperature

(Lobell et al., 2012). Accordingly, selecting cultivars with long-duration and heat-tolerant are needed to adapt to these changes to mitigate the future negative temperature trend impacts in the LP.

5. Conclusions

The observed phenology for winter wheat across the Loess Plateau has changed between 1981 and 2009 because of increased temperature. The warming trend advanced growth stages and reduced phenological durations. The impact of the warming trend on the phenology was partially offset by a continuous, but unspecified introduction of new cultivars with longer thermal time requirement. Sowing dates only affected pre-winter phenology stages. Post-winter phenology stages were driven by the end date of winter dormancy, temperatures until crop maturity, and cultivar characteristics related to temperature responses. Approximately 1/3 of the increased temperature impact on post-winter phenology was compensated by the introduction of new cultivars with altered temperature requirements.

Acknowledgements

This study was supported by National Basic Research Program of China (973 Program, no. 2012CB955304) and National Natural Science Foundation (no. 41105079). The authors acknowledge two anonymous reviewers for their valuable comments.

Appendix A.

Fig. A.1 and Table A.1.

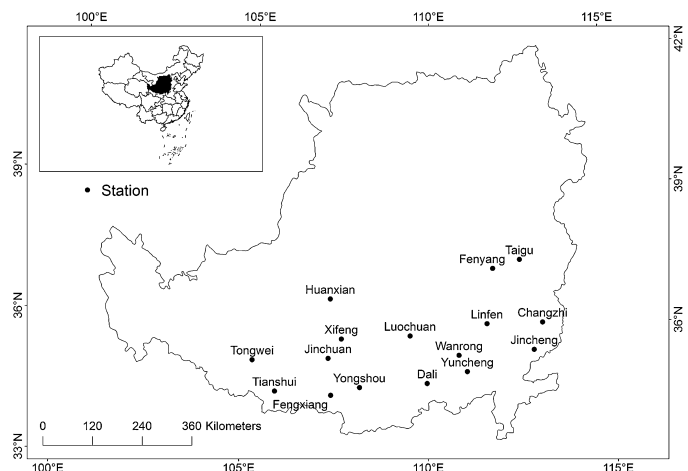
**Fig. A.1.** Map of China with Loess Plateau and locations of agro-meteorological experiment stations.

Table A.1

Winter wheat cultivars at all stations and number of planting years (in brackets).

Stations	Cultivar names of winter wheat (number of planting years)
Taigu	Jinnong-3(1), Jinnong-135(1), Jinnong-849(10), Jing-411(5), TK-882(1), Jingdong-8(2), Nongda-9697(1), Beinong-10(1), Beinongda-948(1), Nongda-86(3), Beinongda-66(2), Lunxuan-987(1)
Fenyang	849(4), 669(1), 584(9), 3039(4), Changzhi-5052(1), Fen-5(1), Fen-596(1), Jing-841(1), Jinmai-75(4), Fen-4846(3)
Changzhi	Jinmai-15(1), 12057(1), Changzhi-648(17), Changzhi-5977(5)
Jincheng	Kangfeng-1(1), Beijing-841(3), 7401(1), 5819(20), Kangyou-1(4)
Linfeng	Jinnan-10(2), Linfen-10(5), 7410(1), 1470(1), K239(7), 514(1), Jinmai-118(3), Jinmai-31(3), Jinmai-61(1), Jinmai-6(1), Jinmai-19(4)
Wangrong	Beijing-10(3), Beijing-12057(7), 7741(2), Jinmai-33(2), Pingyang-27(1), Jinmai-47(14)
Yuncheng	Taishan-1(2), 7410(1), 79-2(2), 78-14(1), 40556(1), 13-1(1), 13-4(1), 1604(4), Shan-229(4), Yunfengzao-21(1), Jilaoshi(3), Jinmai-47(2), Gan6172(1), Yannong-19(4)
Dali	Zhenzhou-761(2), Shicha-8(1), Xiaoyan-6(3), 7852(3), 7859(1), Jinmai-26(3), Xiaoyan-107(4), 8329(1), 8352(1), Qinmai-2(1), Xinong-8(1), Xinong-88(4), Xin-6(3)
Luochuan	Yanan-17(3), Fengkang-13(1), 407(3), Changwu-131(5), Changwu-8918(12), 7801(1)
Yongshou	Fengchan-3(6), Xifu-2(2), Jinmai-21(2), Jinmai-20(1), 83-2(1), 225(1), 229(2), 101(1), Zaoshu-10(1), Shannong8719-14(1), Shanhan-8765(2), Jinmai-54(3), Shannong-791(2), Jinmai-47(4)
Fengxiang	Xiaohong-5(1), Xiaohong-6(2), 699-24(1), Xiannong-151(1), Xiaoyan-6(1), 7859(1), Qinmai-9(1), 44314(1), 8007(1), Xiaoyang-107(3), Aizai(2), Shanmai-897(1), Xiaoyan-107(1), Shanmai-7859(1), Xiaoyan-22(4), Wunong-148(2), Shan-155(1), Xinong-88(1), Xiaoyan22-3(3)
Tongwei	Dongfanghong-3(4), Zhongliang-5(3), Longdong-3(3), Qingshan-821(2), Daxiya(8), Changzhi-2017(1), 157(1), Weinong-4(1), Longjian-19(2), Linyun-125(3), Linyu-125(1)
Huanxian	Qingxuan-27(3), Hongmanzi(2), Xifeng-16(8), Zhuaglang-6(3), Xifeng-19(2), Huandong-1(10), Hongtuzi(1)
Jinchuan	Tianxuan-17(1), Zhongyin-4(1), 7125(2), Hongtuzi(1), Jinmai-20(5), 131(1), 76-89-4(2), 101(5), Lantian(4), Changwu89-3(1), Longyuan-945(1), Changwu-134(1), Pingliang-40(1), 521-7(2), 85-96(1)
Tianshui	Tianxuan-35(1), Tian-763(2), 7289-411(2), 705(1), 7464(20), 7635(3)
Xifeng	Changle-5(4), Qingfeng-1(6), Changwu-131(3), Xifeng-20(9), Xifeng-24(6), Longyu-218(1)

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